

# The M-CATs

SEMINAR 15

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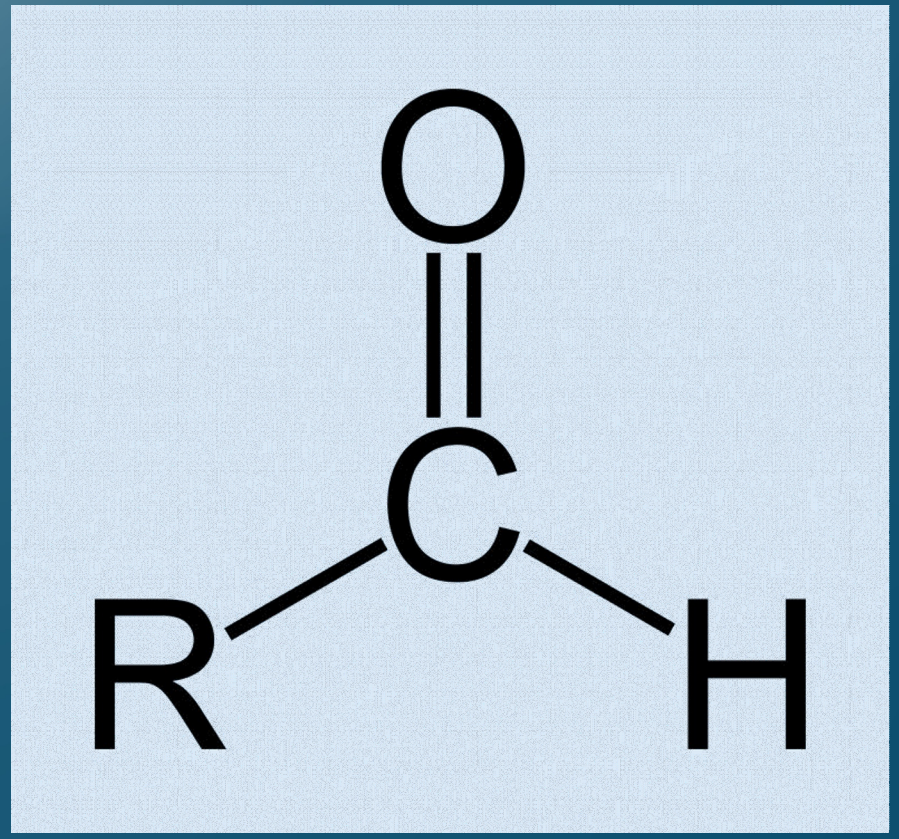


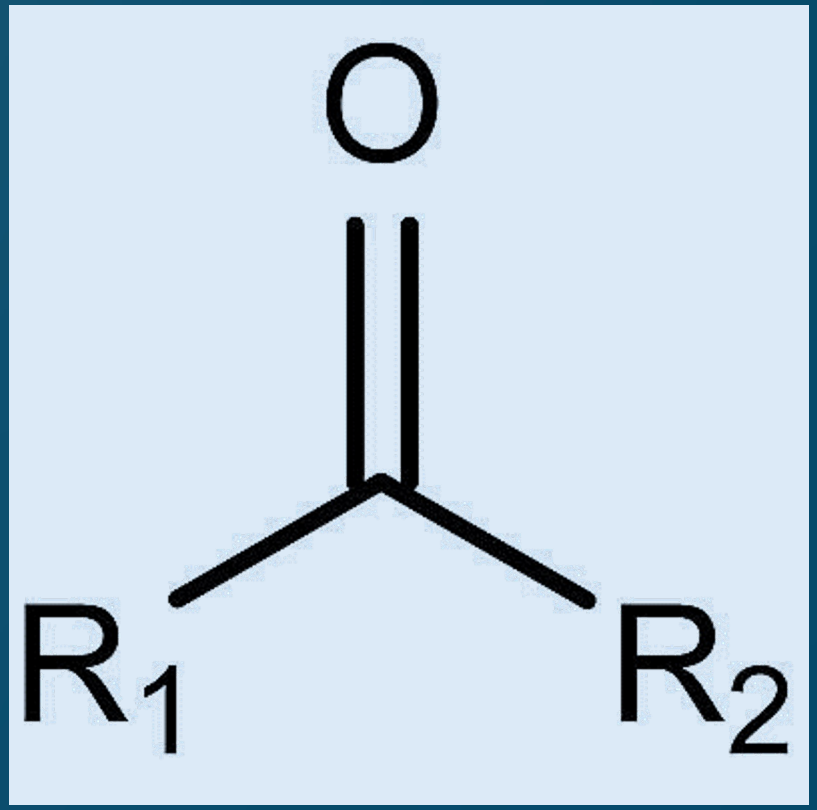

Organic Chemistry Chapters 6, 7

# Aldehydes & Ketones

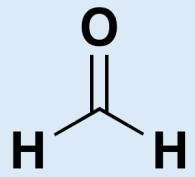


# 6.1. Description & Properties

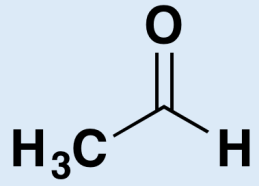


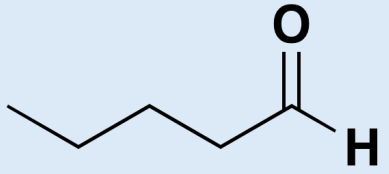
# Nomenclature: Aldehydes



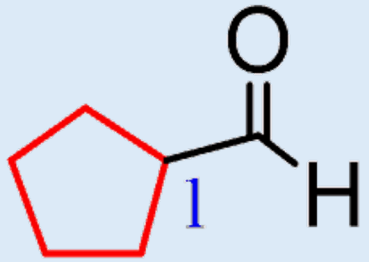

methanal
(formaldehyde)



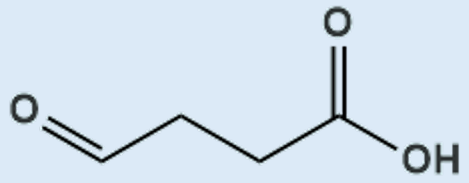

ethanal
(acetaldehyde)




pentanal
(valeraldehyde)

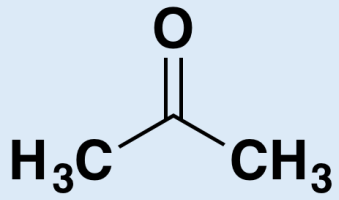



cyclopentanecarbaldehyde

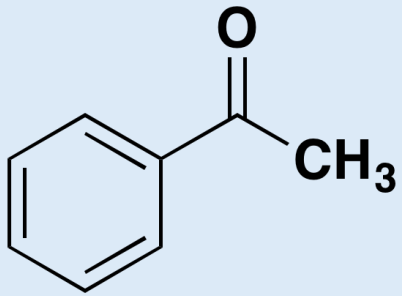



4-oxobutanoic acid

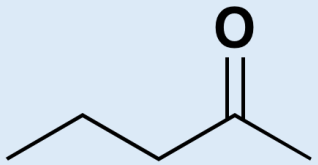
# Nomenclature: Ketones



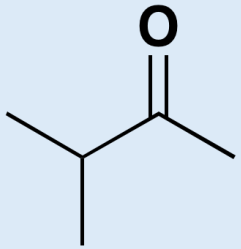

propanone
(acetone)



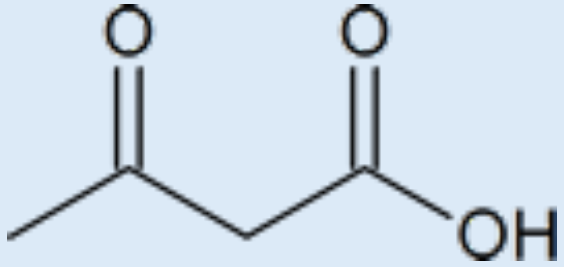

acetophenone
(methyl phenyl ketone)



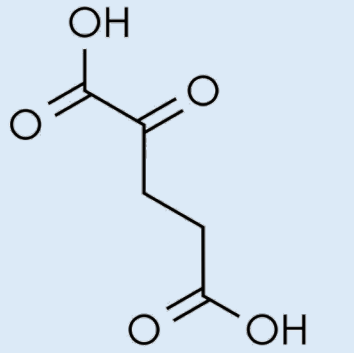

2-pentanone
(methyl propyl ketone)




3-methyl-2-butanone
(methyl isopropyl ketone)

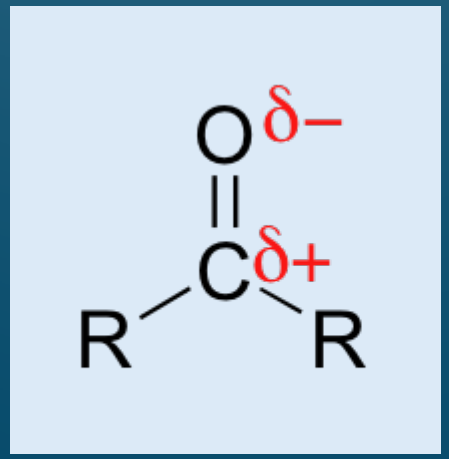



3-oxobutanoic acid




*alpha*-ketoglutaric acid

# Physical Properties



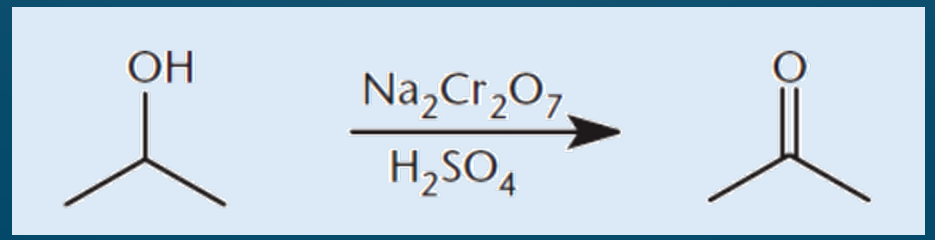
# Question

Q about naming aldehyde and ketone, then estimate BP
A. No—they would reduce all the way to alcohols
B. No—you would end up with unwanted byproducts
C. Yes—works as the reverse of reduction
D. Yes—but only ketones b/c of increased substitution



# Formation



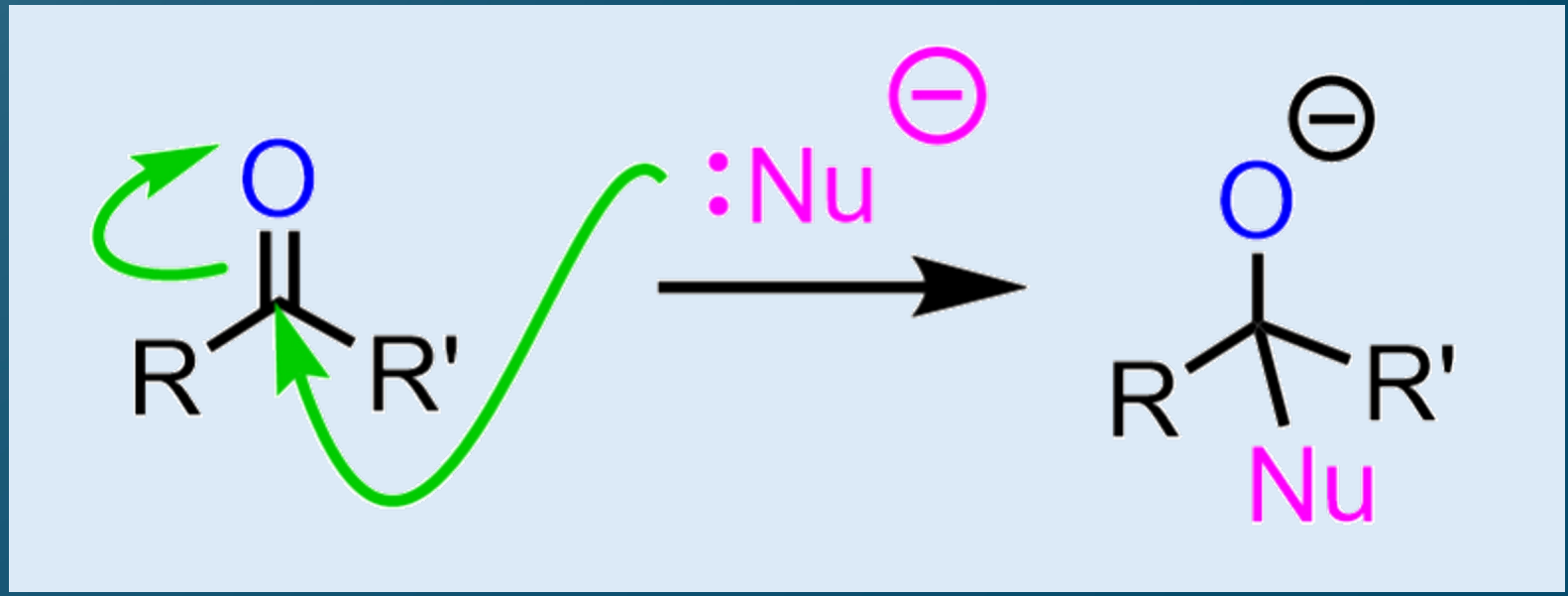
# Question

Can you directly oxidize carboxylic acids into ketones and/or aldehydes?

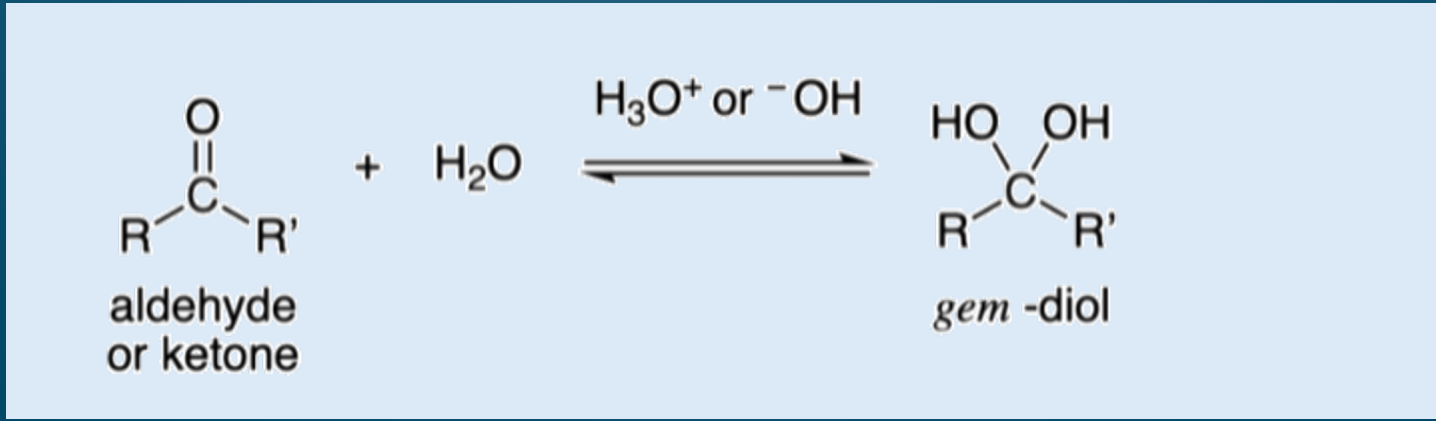
A. No—they would reduce all the way to alcohols
B. No—you would end up with unwanted byproducts
C. Yes—works as the reverse of reduction
D. Yes—but only ketones b/c of increased substitution



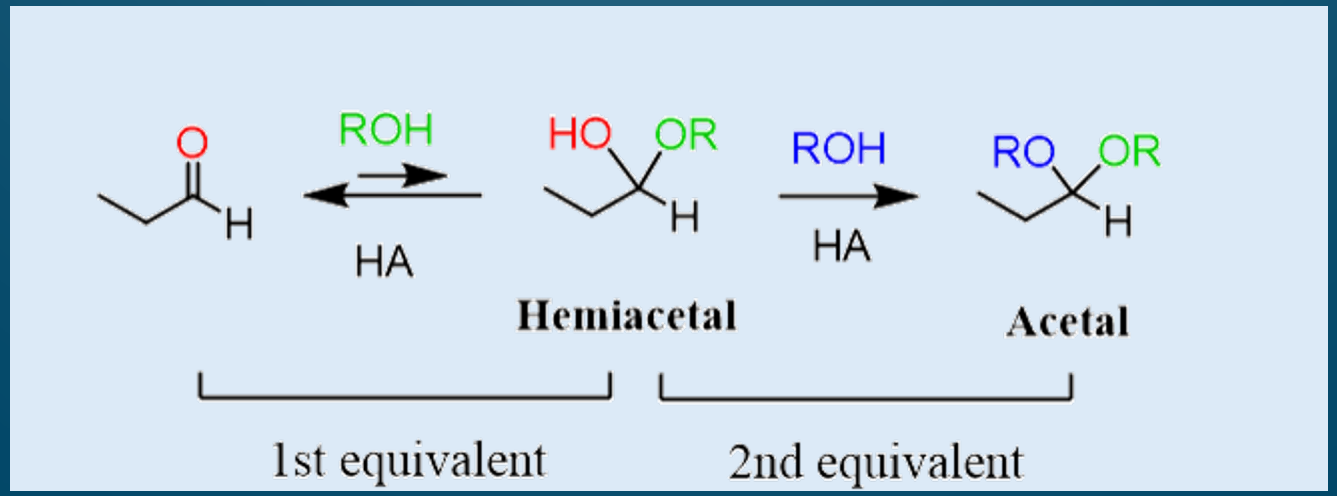
# 6.2. Nucleophilic Addition Reactions

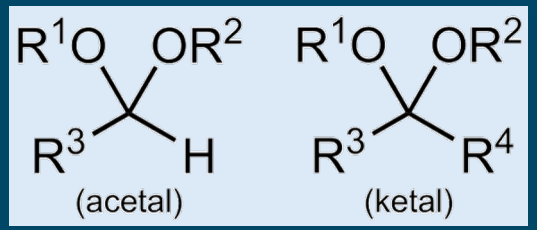


# Hydration

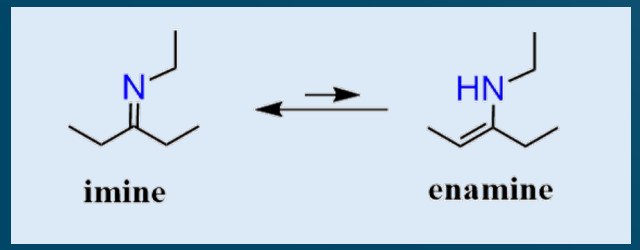


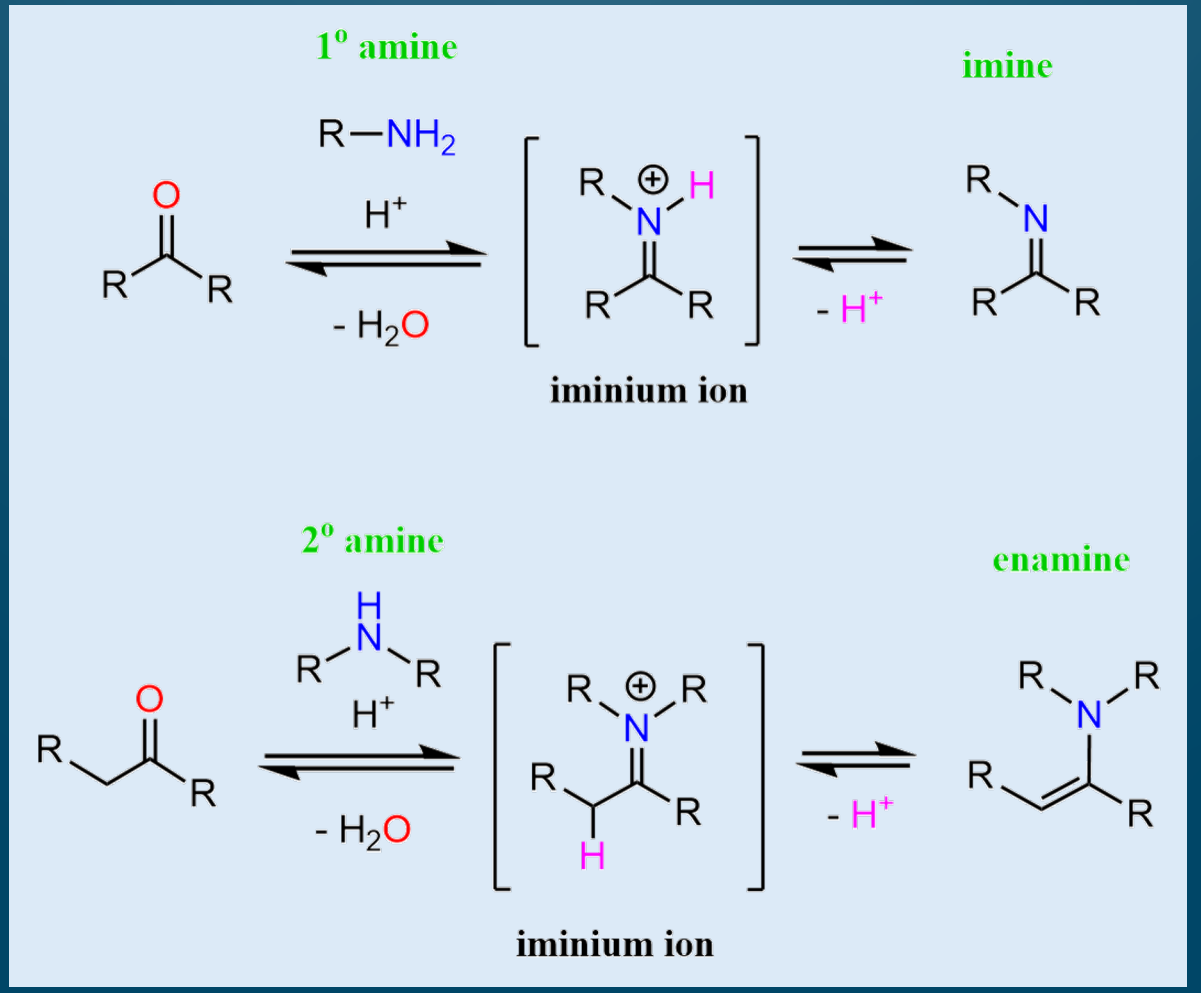
# Hemiacetal/ketal, and acet



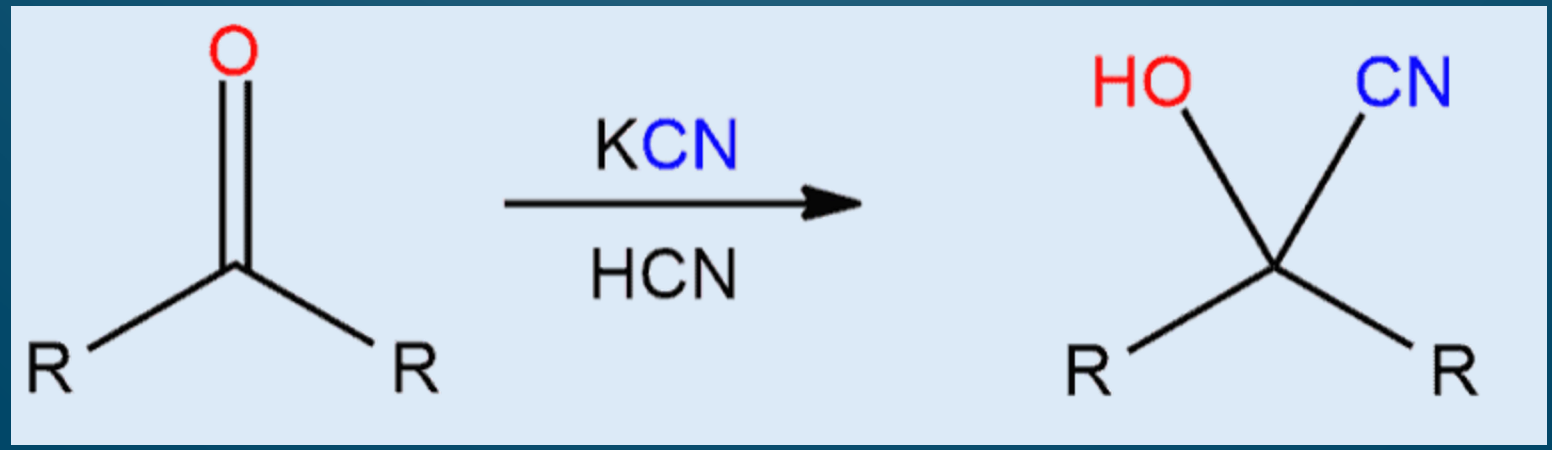



# Imines and Enamines

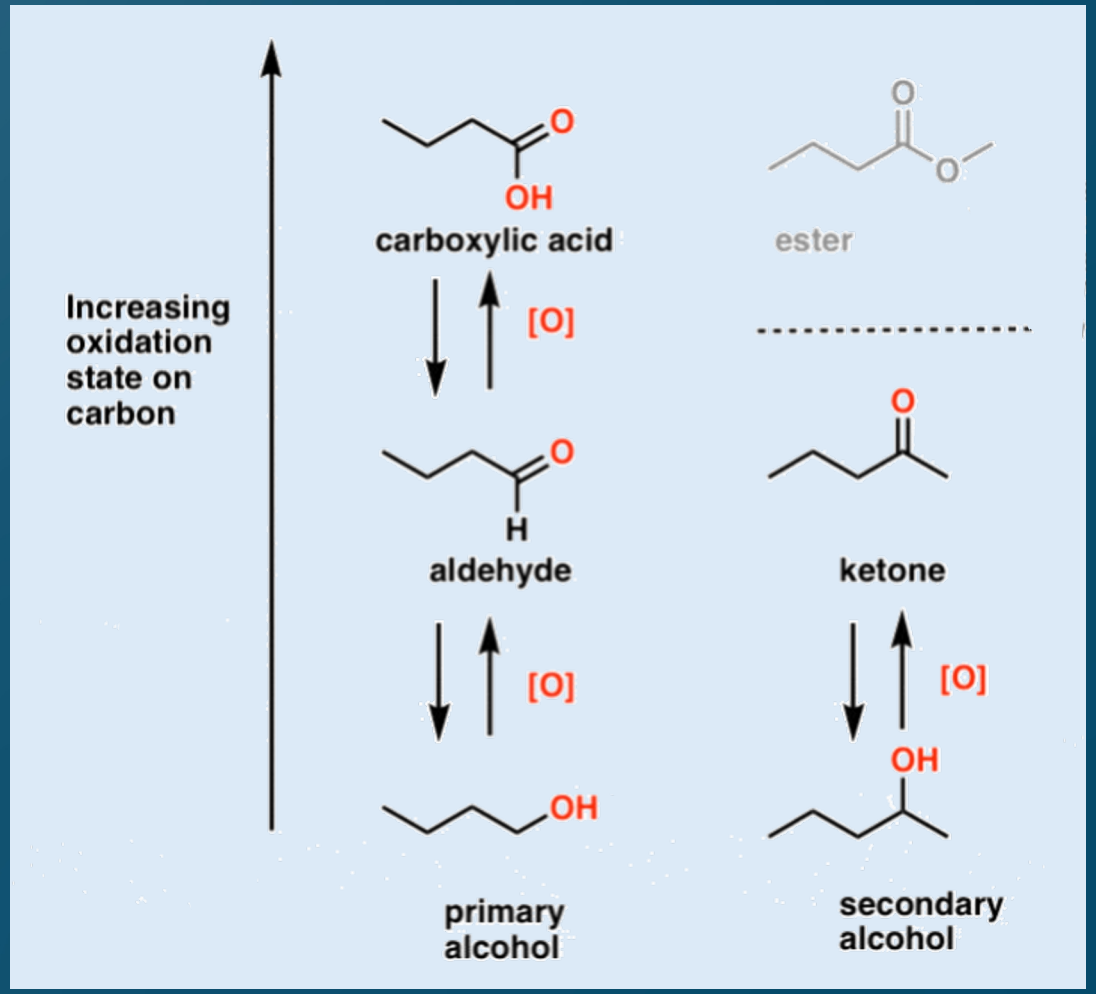


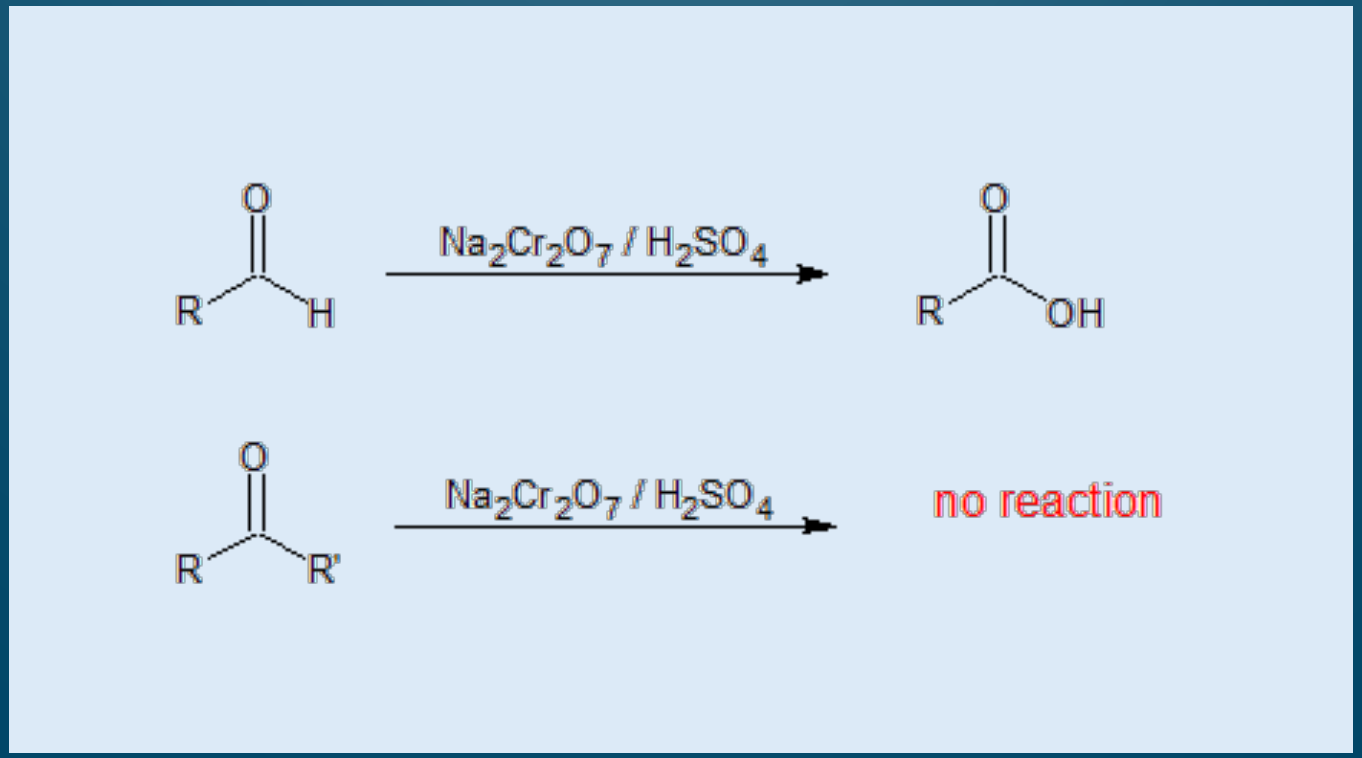
# Cyanohydrins



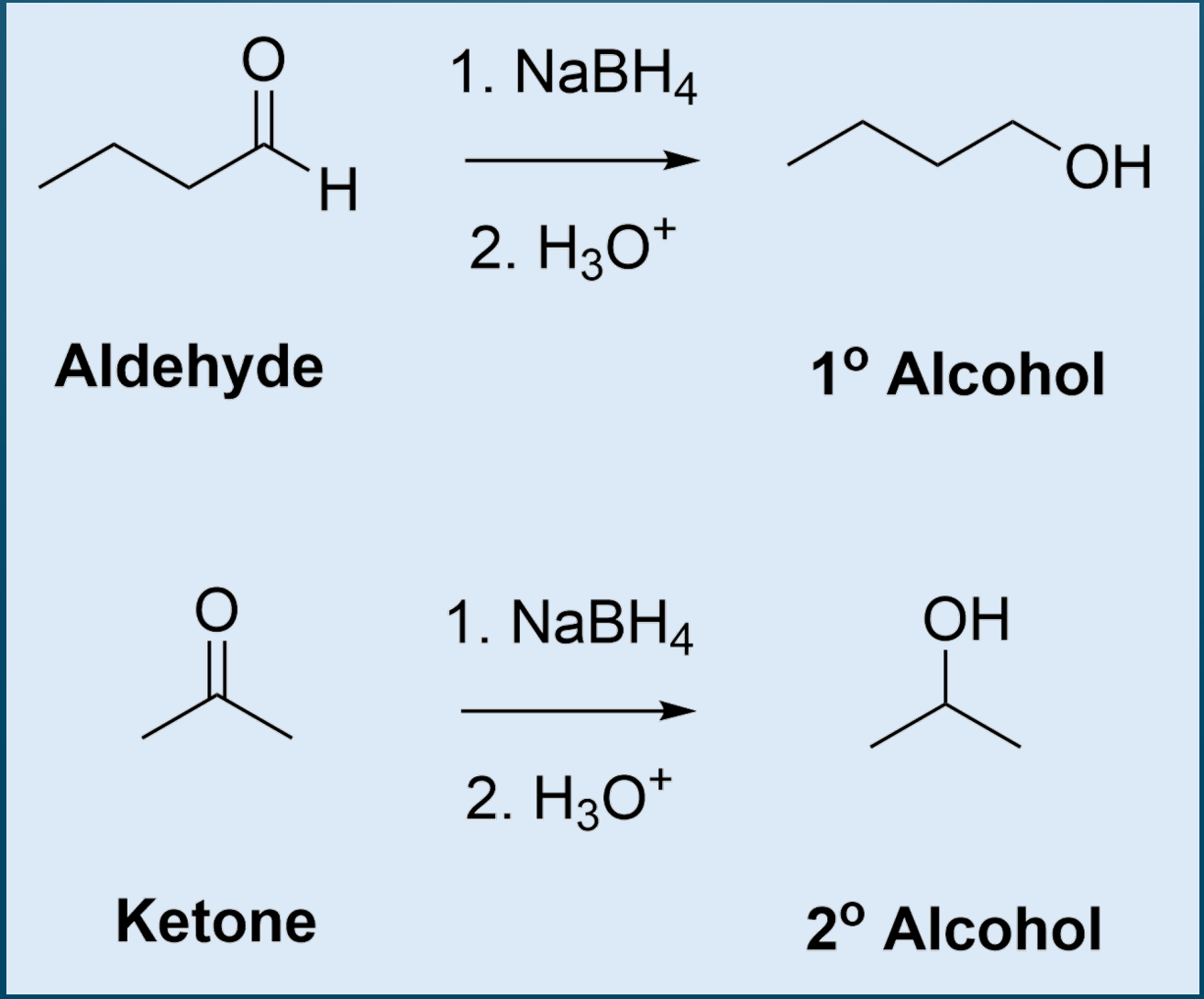
# 6.3. Carbonyl Redox Reactions



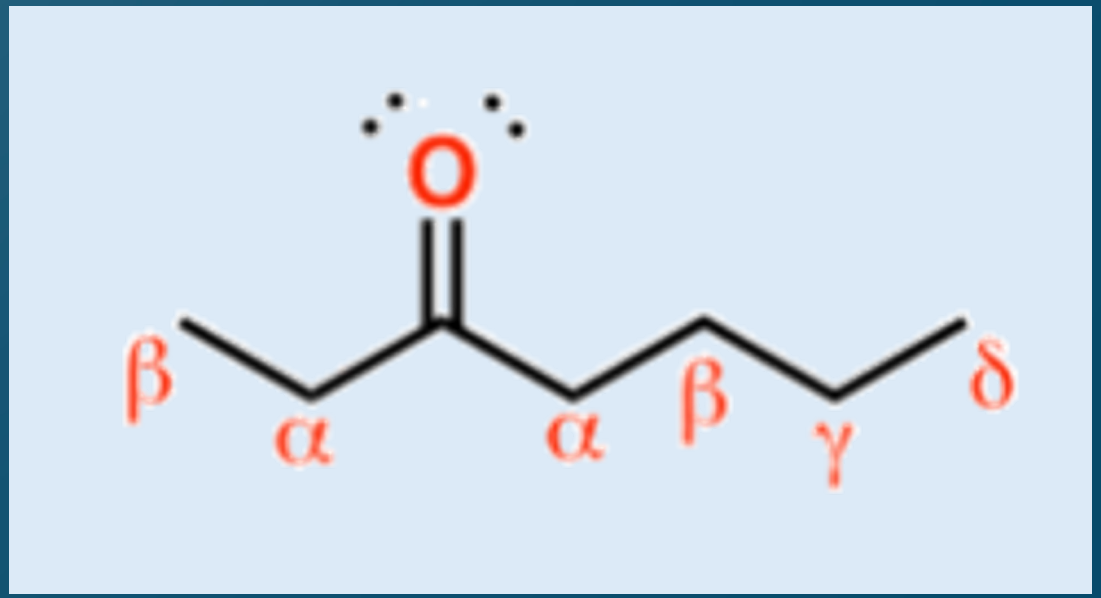
# Oxidation



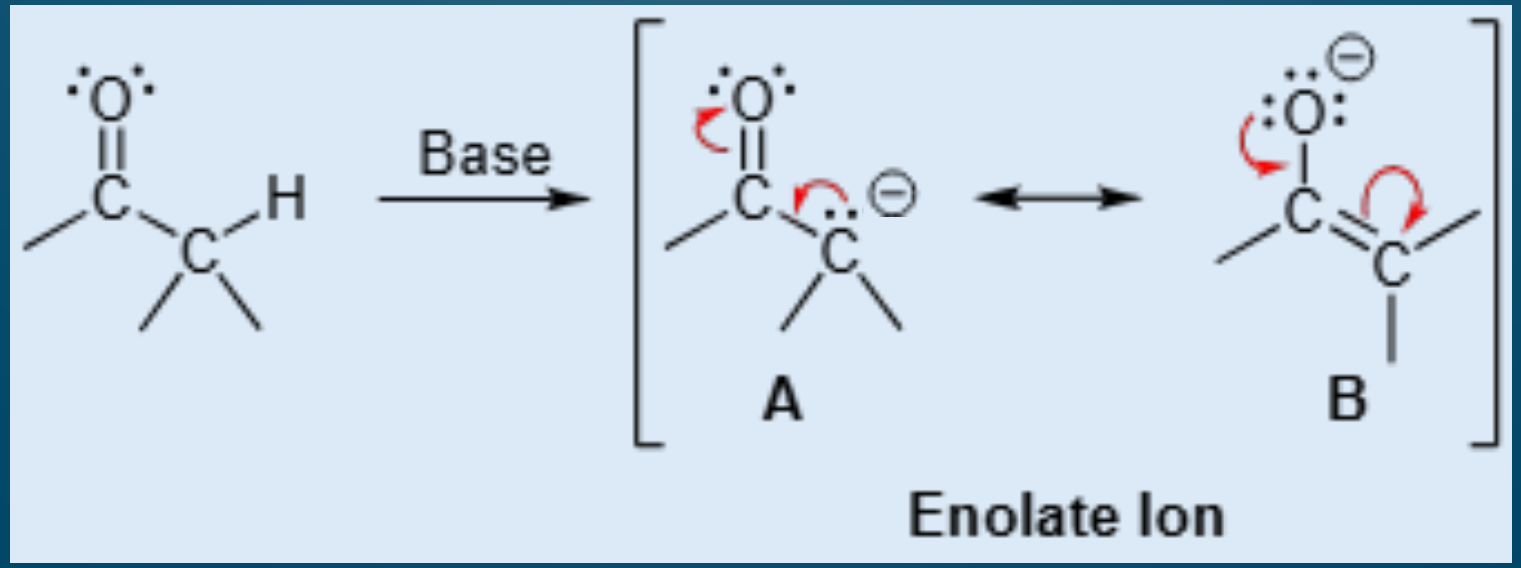
# Reduction


O
H
1. NaBH4
2. H3O+
OH
Aldehyde
1° Alcohol
O
1. NaBH4
2. H3O+
OH
Ketone
2° Alcohol

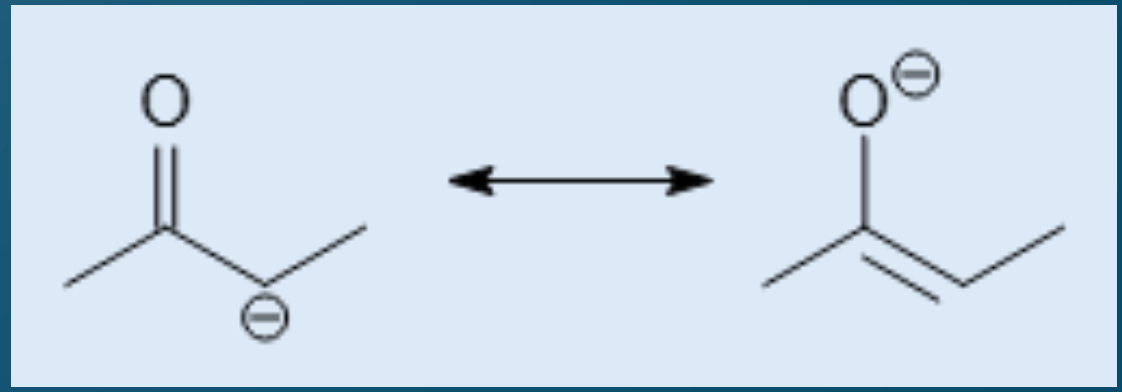
# 7.1. Principles of $\alpha$ -Hydrogens

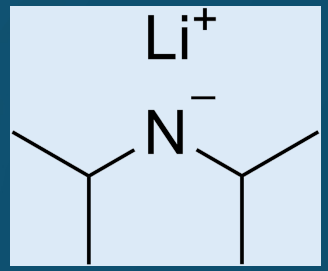


# Acidity



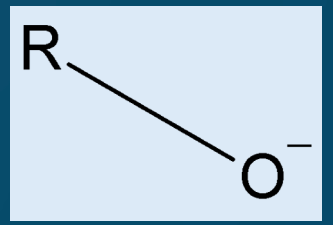
# 7.2. Enolate Chemistry



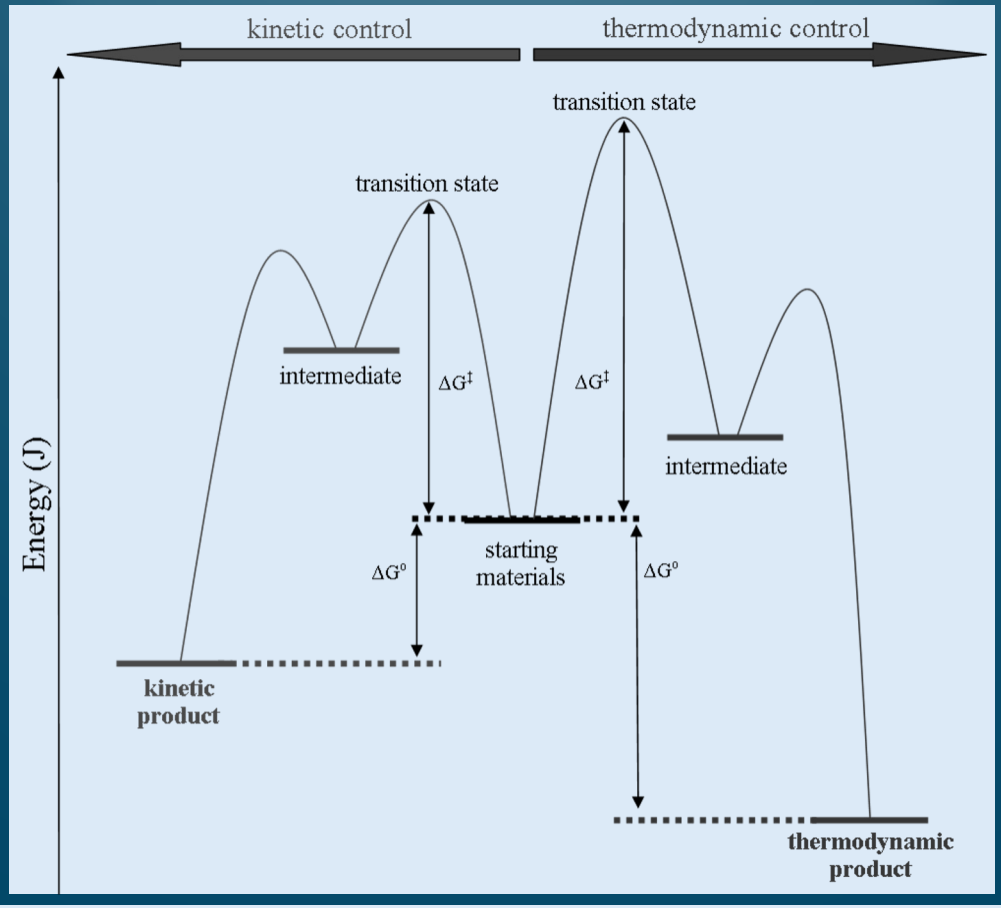



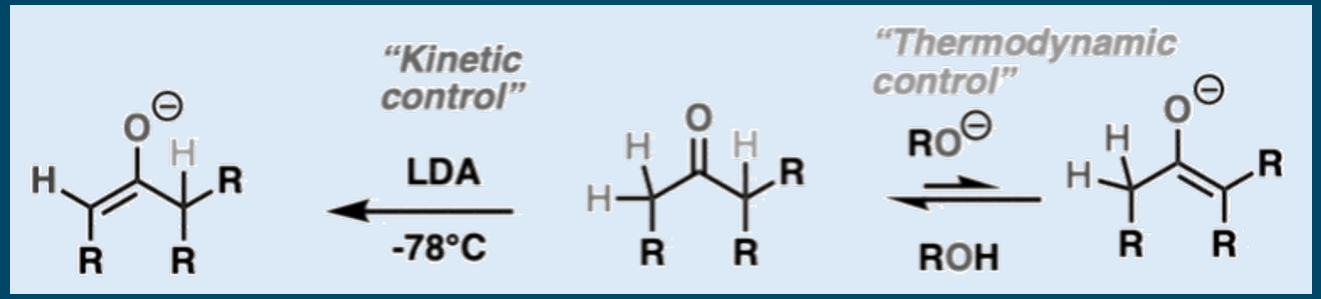




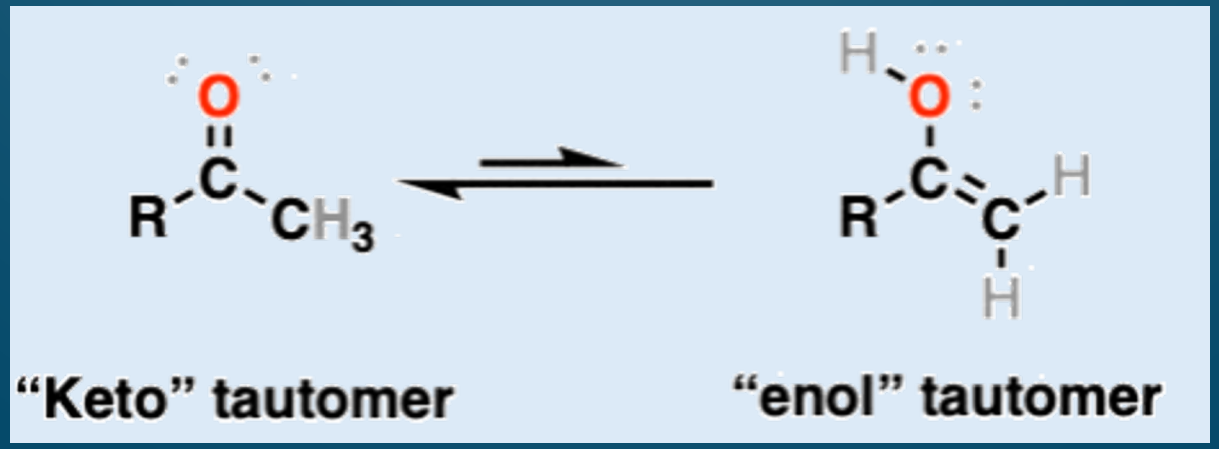


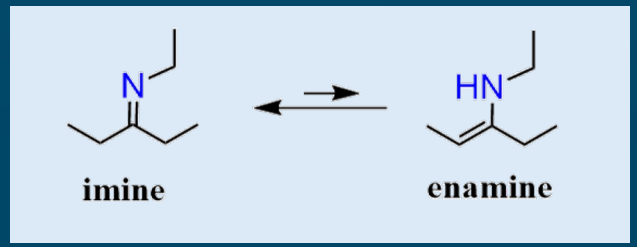
# Thermodynamic/Kinetic Control



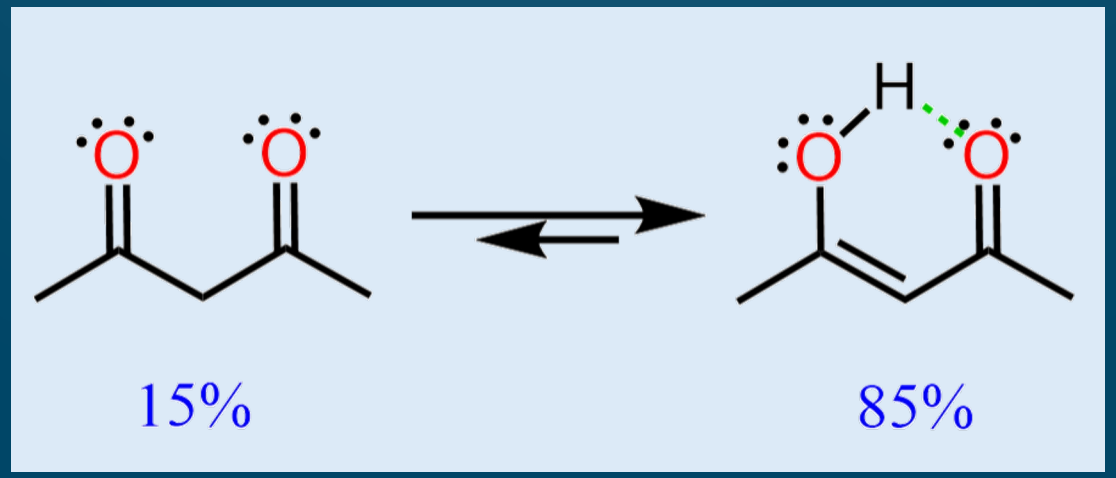
# Tautomerization



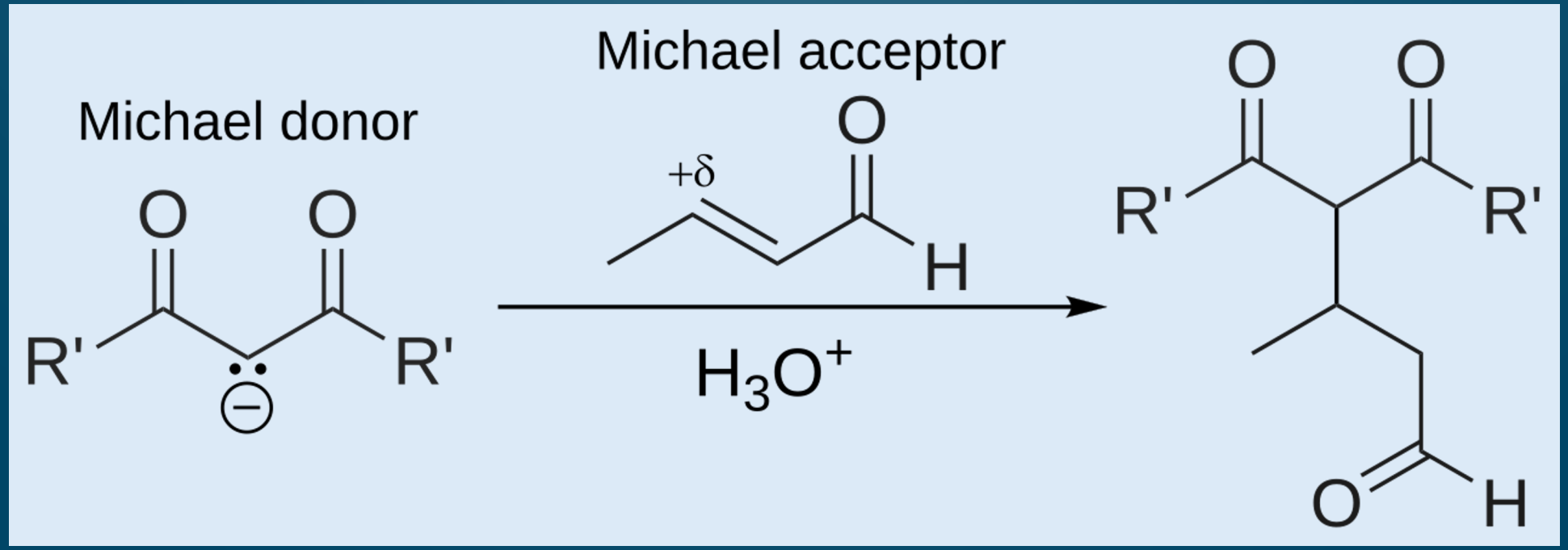



# β-Diketones

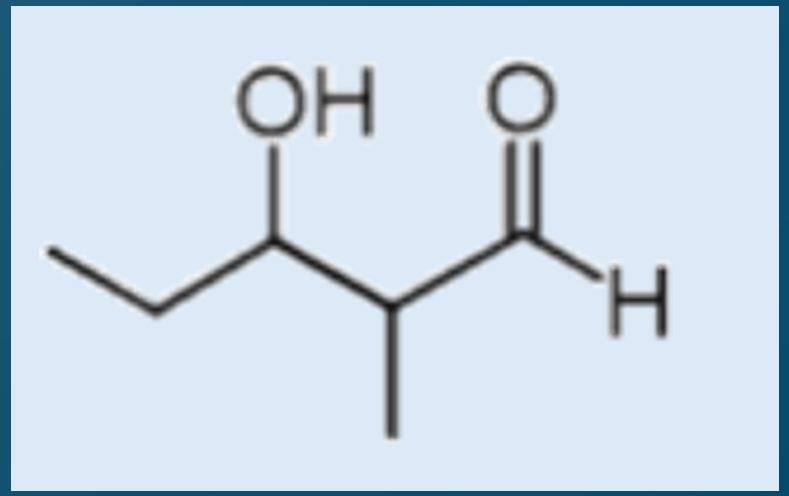


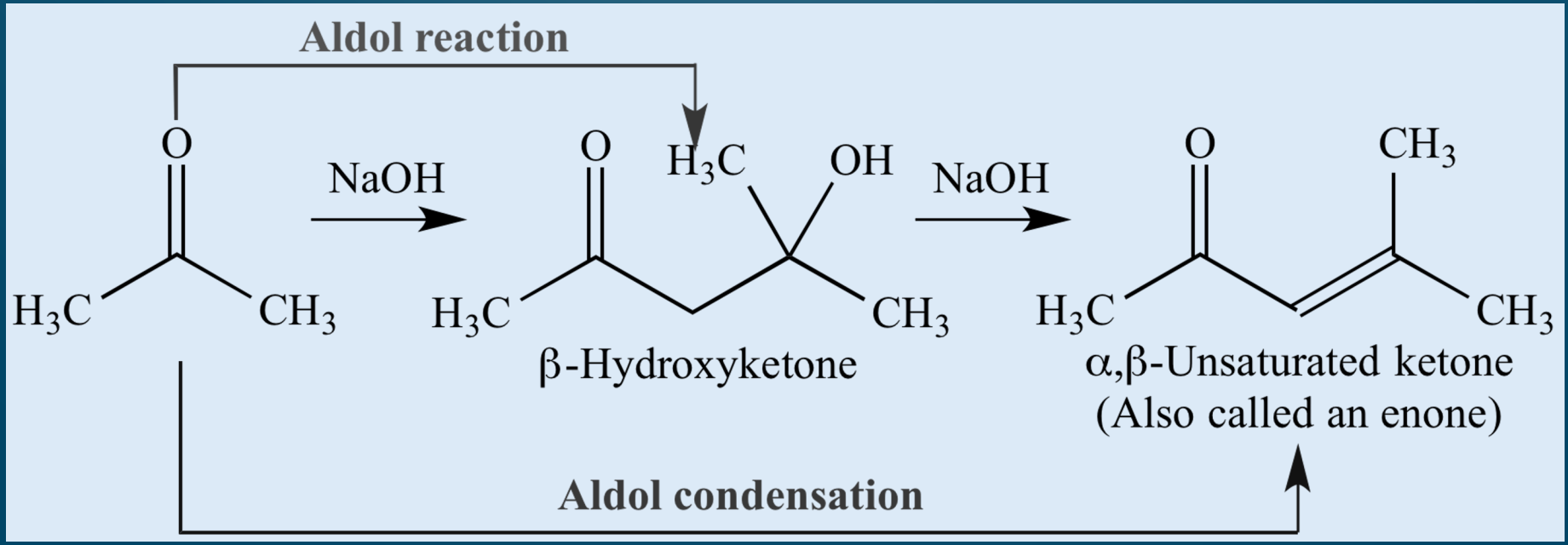
# Michael Addition



# 7.3. Aldols

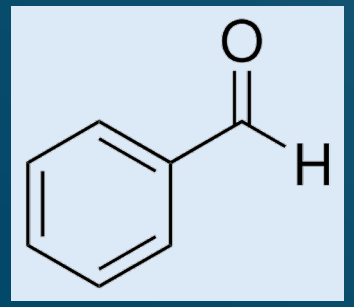


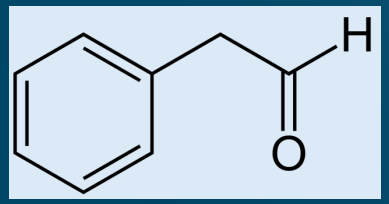
# Aldol Condensation

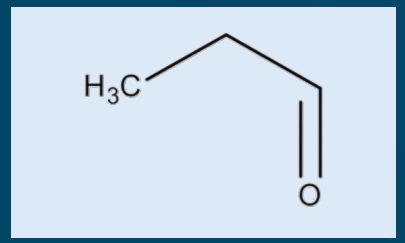


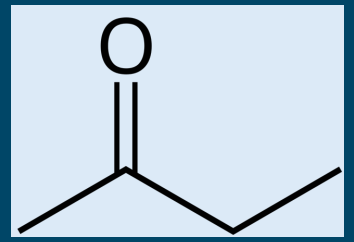
# Question

Which of these compounds will not undergo an aldol reaction?

A. 

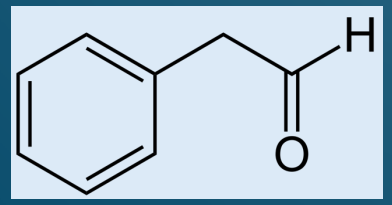

B. 

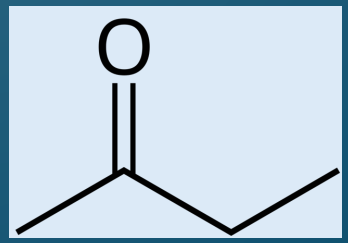

C. 


D. 

# Question

The following two aldols (pictured)are combined in an aldol reaction at a high temperature. How many different aldols may be produced?




A. 0

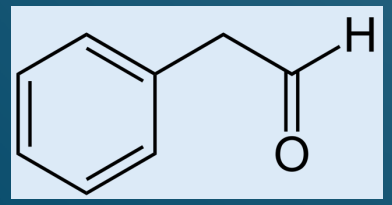
B. 1

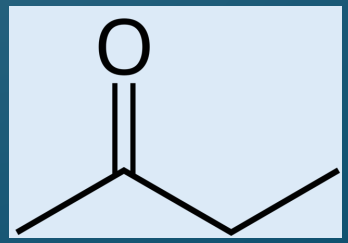
C. 2

D. 4

# Question

The following two aldols (pictured) are combined in an aldol reaction at a *low* temperature. How many different aldols may be produced?




A. 0

B. 1

C. 2

D. 4

# Retro Aldol Reaction

